



**COMSATS University Islamabad, Attock Campus**  
**Department of Computer Science**

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**Program: BCS VIII**

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**SP 2026: \_\_Final Term\_\_ Examination**

**Course: - Parallel and Distributed Computing**

**Time Allowed: - 3hrs**

**Marks: 50**

**Name:- \_\_\_\_\_ Regn. No:- \_\_\_\_\_**

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| <b>Note:- Don't write anything on Question Paper except your Name &amp; Regn. No.</b> |
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Q1. Discuss the different types of Parallel Programming Models. List the possible characteristics of an APIs (Application programming interface). A program that comes with a limitation of a message must be fixed in 16 characters (octets)!

[CLO:5, GA:2,5,9] (10 Marks)

Q2. Break up a long vector into sub-vectors of equal length. Distribute sub-vectors to processes. Let the processes to compute the partial sums. Collect the partial sums from the processes and add them at root node using collective computation operations.

[CLO:5, GA:2,5,9] (15 Marks)

Q3. Parallelize data processing with multiprocessing Queue.

[CLO:5, GA:2,5,9] (5 Marks)

Q4. Write a program to calculate prefix sum  $S_n$  of 'n' numbers on 'n' processes

- Each node has two variables 'a' and 'b'.
- Initially  $a = \text{node id}$
- Each node sends 'a' to the other node.
- 'b' is a variable that receives a sent from another node.

[CLO:5, GA:2,5,9] (10 Marks)

Q5. Write a program in C that calculates the sum of numbers in parallel on different numbers of nodes. Also calculate the execution time using GPU.

[CLO:5, GA:2,5,9] (5 Marks)

Q4. Viva. (5 Marks)

***[CLO-5 Develop an application using concepts of distributed and parallel computing in a team environment.]***

